

CONTACT:

Jules Schleinitz

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EDUCATION:

California Institute of Technology, Postdoc., 2022 – Present

École Normale Supérieure – PSL, Ph. D. Molecular Chemistry, 2022

Sorbonne Université, M. S. Theoretical Chemistry, 2019

Université Paris Saclay, M. S. Teaching Chemistry (Agrégation), 2017

École Normale Supérieure, B. S. Chemistry, 2015

Lycée Thiers, PSCI/PC* admission to ENS Ulm, 2014

PROFESSIONAL SUMMARY:

I transitioned from a theoretical background in mathematics, physics, and theoretical chemistry to experimental organic chemistry with a focus on mechanistic studies. I have five years of experience in applying machine learning to organic reactions yield and selectivity predictions, fostering collaborations between experimental and computational teams. I have expertise in Python, active learning, reactivity featurization, DFT, experimental mechanistic studies, organic synthesis, and high throughput experimentation. Three years of chemistry teaching experience.

RESEARCH POSITIONS:

California Institute of Technology, Postdoc, 2022 – Present with Prof. Sarah Reisman.

- Active learning to designing small datasets to perform accurate predictions on complex molecules.
- Dataset design for building of models of reactivity.
- Building models of reactivity across ligands classes for Nickel cross electrophiles couplings (Ni-CEC).
- Prediction of matching electrophiles across heterocycles classes in Ni-CEC.
- Development of a GC-MS/FID high throughput experimentation platform (with Dr. Scott Virgil)

École Normale Supérieure – PSL, Ph. D. Molecular Chemistry, 2019-2022 with Dr. Laurence Grimaud.

- Theoretical and experimental mechanistic study of a photo-catalyzed *N*-oxide deoxygenation.
- Theoretical and experimental mechanistic study of Pd-mediated nitro cross-coupling.
- Electrochemical studies of Nickel complexes.
- Reaction yield prediction from academic literature data.

Aqemia, with Dr. Maximilien Levesque (CEO of Aqemia), 2020.

- Implementation of scoring functions to evaluate the *synthetizability* of drug-like molecules.

École Polytechnique, with Dr. Grégory Nocton, 2018

- Synthesis and characterization of divalent lanthanide dimers and sandwich single-molecule magnets

Technische Universität Berlin, with Pr. Martin Kaupp, 2016

- Analysis of the inverse trans influence on ^1H NMR hydride shifts in U^{VI} complexes with relativistic DFT.

École Polytechnique, with Dr. Gilles Frison, 2015

- Description of the coordination properties of *N*-heterocyclic mesoionic carbens with DFT.

École Normale Supérieure Paris, with Dr. Agathe Espagne, 2015

- Synthesis and spectroscopic characterization of the photoswitchable protein Padron.

PUBLICATIONS: *corresponding author, †equal contributions.

- Designing Target-specific Data Sets for Regioselectivity Predictions on Complex Substrates**
J. Schleinitz, A. Carretero-Cerdan[†], A. Gurajapu[†], Y. Harnik, G. Lee, A. Pandey, A. Milo*, S. E. Reisman*. *J. Am. Chem. Soc.* **2025**, 147, 9, 7476–7484.
- Oxidative Addition of Nitro-Perylenediimide onto Palladium(0): A theoretical and experimental study**
J. Schleinitz, C. Chinchilla-Garzon, A. Perfetto, E. Escoudé, A. Makhouloutah, G. Gontard, M. R. Vitale, A. Goujon, F-X. Felpin, L. Binet, I. Ciofini, P. Hudhomme*, L. Grimaud.* *Organometallics*. **2025**, 44, 1, 29–35.
- Site-Selective Radical Aromatic C–H Functionalization of Alloxazine and Flavin through Ground-State Single Electron Transfer**
A. Das, O. Charpentier, C. Hessin, J. Schleinitz, D. Pianca, N. Le Breton, S. Choua, L. Grimaud, C. Gourlaouen, M. Desage-El Murr*. *Angewandte Chemie International Edition*. **2024**, e202403417.
- Dataset Design for Building Models of Chemical Reactivity**
P. Raghavan, B. Haas[†], M. Ruos[†], J. Schleinitz[†], A. Doyle, S. Reisman, M. Sigman, C. Coley*. *ACS Central Science*. **2023**, 9, 12, 2196–2204

6. **Assessing the Extent of Potential Inversion by Cyclic Voltammetry: Theory, Pitfalls, and Application to a Nickel Complex with Redox-Active Iminosemiquinone Ligands**
C. Hessin, **J. Schleinitz**, N. Le Breton, S. Choua, L. Grimaud, V. Fourmond, M. Desage-El Murr*, and C. Léger*, *Inorg. Chem.* **2023**, 62, 8, 3321-3332.
5. **Machine Learning Yield Prediction from NiCOLit, a Small-Size Literature Data Set of Nickel Catalyzed C–O Couplings**
J. Schleinitz^{†*}, M. Langevin^{†*}, Y. Smail, B. Wehnert, L. Grimaud*, R. Vuilleumier*. *J. Am. Chem. Soc.* **2022**, 144, 32, 14722–14730.
This article was selected for **JACS Spotlight** (*J. Am. Chem. Soc.* **2022**, 144, 33, 14961)
4. **A single bioinspired hexameric nickel catechol-alloxazine catalyst combines metal and radical mechanisms for alkene hydrosilylation**
A. Das, **J. Schleinitz**, L. Karmazin, B. Vincent, N. Le Breton, G. Rogez, A. Guenet, S. Choua, L. Grimaud, M. Desage-El Murr*. *Chem. Eur. J.* **2022**, 28, e2022005.
3. **A hybrid bioinspired catechol-alloxazine triangular nickel complex stabilizing protons and electrons**
A. Das, H. Jobelius, **J. Schleinitz**, S. Gamboa-Ramírez, G. Creste, G. Kervern, J. Raya, N. Le Breton, A. Guenet, Z. Boubegiten, L. Grimaud, M. Orio, G. Rogez, P. Hellwig, S. Choua, S. Ferlay, M. Desage-El Murr*. *Inorg. Chem. Front.*, **2021**, 8, 5286-5298.
2. **Metal-Free Deoxygenation of Amine N-Oxides: Synthetic and Mechanistic Studies**
W. Lecroq[†], **J. Schleinitz**[†], M. Billoue, A. Perfetto, A-C. Gaumont, J. Lalevée, I. Ciofini, L. Grimaud, S. Lakhdar., *ChemPhysChem*, **2021**, 22, 1237
1. **Bis-Cyclooctatetraenyl Thulium(II): Highly Reducing Lanthanide Sandwich Single-Molecule Magnets**
J. Moutet, **J. Schleinitz**, L. La Droitte, M. Tricoire, F. Pointillart, F. Gendron, T. Simler, C. Clavaguéra, B. Le Guennic, O. Cador, G. Nocton*. *Angewandte Chemie International Edition*, **2021**, 60 (11), 6042-6046.

ORAL PRESENTATIONS:

GRC – Organic Reaction and Processes, Bryant University, USA. 2024

Active Learning-Guided Dataset Design Enables the Prediction of Regioselectivity for Late-Stage C-H Functionalization on Complex Molecules. Poster

AI=Science: Strengthening the Bond Between the Sciences and AI, The Simons Institute, UC Berkeley, USA. 2024

Active Learning-Guided Dataset Design Enables the Prediction of Regioselectivity for Late-Stage C-H Functionalization on Complex Molecules”. Oral Communication

C-CAS – Annual Meeting, UCLA, USA. 2024

Active Learning-Guided Dataset Design Enables the Prediction of Regioselectivity for Late-Stage C-H Functionalization on Complex Molecules”. Poster

ACS Fall 2023, Machine Learning for Organic Chemistry, San Francisco, USA. 2023

Machine Learning Yield Prediction from NiCOLit, a Small-Size Literature Data Set of Nickel Catalyzed C–O Couplings. Oral Communication

EPFL, AI in Chemistry and beyond. Lausanne, Switzerland. Online. 2023

Predicting Regioselectivity on Complex Molecules. Oral Communication

UCLA, Doyle Group Seminar. Los Angeles, USA. 2022

Machine Learning Yield Prediction from NiCOLit, a Small-Size Literature Data Set of Nickel Catalyzed C–O Couplings.

Hong-Kong University, Chemistry Department Seminar, Hong-Kong. Online. 2022

Machine Learning Yield Prediction from NiCOLit, a Small-Size Literature Data Set of Nickel Catalyzed C–O Couplings.

Institut Curie, 6th Journées de la Montagne Sainte Geneviève, Paris, France. 2022

Machine Learning Yield Prediction from NiCOLit, a Small-Size Literature Data Set of Nickel Catalyzed C–O Couplings.

LEADERSHIP POSITION:

Organization of Machine Learning and Chemistry Literature Sessions: “ML+Chem”, Caltech, Pasadena, CA.

Bi-weekly hybrid seminars in which I invite world leaders from industry and academia to present their latest developments in the field. Invited people were working at UCLA, Caltech, NASA-JPL, Terray Therapeutics, Aqemia, Atinary, EPFL, Ohio State University, and CNRS among others.

Reviewer:

Digital Discovery

Journal of Chemical Information and Modelling

TEACHING POSITION:

Agrégé Préparateur, ENS, Paris, France. 2019-2022

Teaching *How to teach Chemistry* to a one-year curriculum class of ~10 future teachers in chemistry (equivalent to a master's degree). Four hours sessions including the presentation of a lecture or a practical demonstration by a student, the analysis of the presentation highlighting positive practices and potential improvements. About 15 sessions/year.

Recruiter of ENS chemistry students, ENS, Paris, France. 2019-2022

Writing and proctoring experimental evaluation sessions (~four weeks/year).

Examination conception and corrections (~three weeks/year).

Organic Chemistry Instructor, ENS, Paris, France. 2019-2022

14 sessions of two hours/year for a class of ~30 students. Including lectures, exercises, and continuous evaluations.

Electrochemistry Teaching Assistant, ENS, Paris, France. 2019-2022

Seven sessions of 1.5 hours/year (theoretical courses) & three sessions of three hours practical sessions.

Organic Lab Teaching Assistant, ENS, Paris, France. 2019-2022

Three sessions of six hours/year (theoretical courses) & final evaluation

Physics Oral Examiner, undergraduate students: hour-long oral examination of a group of three students.

Lycée Condorcet, Paris, France. Two hours per week. 2017-2018

Lycée Louis Le Grand, Paris, France. Two hours per week. 2017-2018

MENTORING EXPERIENCE:

California Institute of Technology.

- Computer science undergraduate student (co-mentored with Dr. Raul Astudillo): Transposing Bayesian optimization efficiency to active learning strategies for molecular DFT properties prediction. *Summer 2024.*
- Undergraduate students: Leveraging HTE to quantify reaction substrate-scope for Ni-catalyzed cross electrophile couplings. 2 students for 2 months. *Summer 2024.*
- Undergraduate students: Predicting the regioselectivity of dioxirane C–H oxidations. (two students for two months). *Summer 2023*
- Undergrad research thesis: Benchmarking reaction yield prediction across reaction representations (one student for four months). *Spring-Summer 2023*
- Undergrad research thesis: Few Shot Learning to Transfer Reaction Yield Prediction Models (one student for two months). *Spring 2023*
- Graduate students (one student): Dataset design for regioselectivity predictions. 2022- *Present.*

École Normale Supérieure – PSL,

- Masters’ internship: Machine learning prediction of Lewis acidities. Five months. 2022
- Masters’ internship: Cheminformatics for modeling literature data. Two students for five months. 2021
- Masters’ internship: DFT and electrochemistry for the understanding of a Suzuki-type Palladium nitroarene coupling. Five months. 2020
- Five Bachelor students for internships varying from two weeks to three months on electrochemistry projects and reactivity modeling. 2019-2022

LANGUAGES:

French (native speaker), English (fluent), Spanish (conversant), German (beginner), and Russian (beginner).